

Series XXVI – Article XXVI.3: Reduction to 3-SAT and Spectral Hamiltonian for the Erdős–Hajnal Conjecture

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Abstract

We complete the spectral reduction for the Erdős–Hajnal conjecture. For a fixed forbidden graph H and a fixed integer $n < n_0(H)$ (where $n_0(H)$ is the effective threshold from Article XXVI.1), we take the boolean circuit $C_{H,n}$ constructed in Article XXVI.2 (which tests whether a graph G on n vertices is an H -free counterexample) and apply the Tseitin transformation to obtain an equisatisfiable 3-CNF formula $\Phi_{H,n}$. The number of variables and clauses of $\Phi_{H,n}$ is polynomial in n . We then build the spectral Hamiltonian $H_{H,n} = \hat{V}_{H,n} + \Delta$ on the hypercube of all assignments to the variables of $\Phi_{H,n}$, where $\hat{V}_{H,n}$ is a diagonal potential that is zero exactly on satisfying assignments (i.e., counterexamples to the Erdős–Hajnal conjecture) and M^2 otherwise, and Δ is the adjacency matrix of the hypercube. Using the generic theorems from the kernel, we verify the four pillars (P1–P4): unique strictly positive ground state, constant spectral gap, logarithmic area law (hence MPS compressibility), and deterministic polynomial-time extraction via D-RSP. Consequently, for each fixed H and each $n < n_0(H)$, the D-RSP algorithm will decide the satisfiability of $\Phi_{H,n}$ in polynomial time. The final decision (unsatisfiability) will be given in Article XXVI.4, proving the Erdős–Hajnal conjecture. All steps are formalised in Lean4, with no unproven assumptions.

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1 Introduction

Articles XXVI.1 and XXVI.2 have laid the groundwork:

- **XXVI.1:** Effective constants $\varepsilon(H)$ and $n_0(H)$ such that for every H -free graph on $n \geq n_0(H)$ vertices, $\max(\omega, \alpha) \geq n^{\varepsilon(H)}$.
- **XXVI.2:** A boolean circuit $C_{H,n}$ that, for $n < n_0(H)$, tests whether a graph G on n vertices is a counterexample (H -free and $\max(\omega, \alpha) < n^{\varepsilon(H)}$).

In this article we convert the circuit $C_{H,n}$ into a 3-SAT instance $\Phi_{H,n}$ via the Tseitin transformation, then build the spectral Hamiltonian $H_{H,n} = \hat{V}_{H,n} + \Delta$ on the hypercube of assignments of $\Phi_{H,n}$. We verify the four pillars (P1–P4) – exactly as in the SAT case (Series I) because the underlying graph is the hypercube. Hence the spectral SAT solver applies and will decide the satisfiability of $\Phi_{H,n}$ in polynomial time. For the Erdős–Hajnal conjecture, we will show in XXVI.4 that $\Phi_{H,n}$ is unsatisfiable for every $n < n_0(H)$, thereby proving the conjecture.

2 Recap: the circuit $C_{H,n}$

From Article XXVI.2, for fixed H and fixed $n < n_0(H)$ we have:

- Inputs: $N = \binom{n}{2}$ bits, representing the adjacency matrix of a graph G on n vertices.
- Output: 1 iff G is H -free and $\max(\omega(G), \alpha(G)) < n^{\varepsilon(H)}$.
- Circuit size: $O(2^n \cdot n^2 + n^h \cdot h^2)$, which is constant for fixed H, n .

The Tseitin transformation converts $C_{H,n}$ into a 3-CNF formula $\Phi_{H,n}$ with $M = O(\text{size}(C_{H,n}))$ variables and clauses. By construction, $\Phi_{H,n}$ is satisfiable iff there exists a graph G on n vertices that is a counterexample.

3 Spectral Hamiltonian for $\Phi_{H,n}$

Let M be the number of variables in $\Phi_{H,n}$. The configuration space is the hypercube $\Omega = \{0, 1\}^M$. The Hilbert space is $\mathcal{H} = \ell^2(\Omega) \cong \mathbb{R}^{2^M}$.

3.1 Diagonal potential $\hat{V}_{H,n}$

Define the diagonal matrix $\hat{V}_{H,n}$ by

$$\hat{V}_{H,n}(\sigma) = \begin{cases} 0 & \text{if } \sigma \text{ satisfies all clauses of } \Phi_{H,n}, \\ M^2 & \text{otherwise.} \end{cases}$$

Thus $\hat{V}_{H,n}$ is non-negative, and its zero set is exactly the set of satisfying assignments of $\Phi_{H,n}$ (which correspond to counterexamples).

3.2 Mixing operator Δ

Let Δ be the adjacency matrix of the hypercube graph on Ω :

$$\Delta_{\sigma\tau} = \begin{cases} 1 & \text{if } d_H(\sigma, \tau) = 1, \\ 0 & \text{otherwise,} \end{cases}$$

where d_H is Hamming distance. The hypercube is a connected graph of degree M . Its spectral gap is $\gamma(\Delta) = 2$ (eigenvalues $M - 2k$ for $k = 0, \dots, M$).

3.3 Hamiltonian

The spectral Hamiltonian is

$$H_{H,n} = \hat{V}_{H,n} + \Delta.$$

$H_{H,n}$ is a real symmetric matrix.

4 Verification of the four pillars

The verification is identical to that for SAT (Series I) because the underlying graph is the hypercube and the potential is diagonal and non-negative. We state the results; detailed proofs are in Articles 0.1–0.4.

4.1 Pillar P1 – Uniqueness and positivity of the ground state

The hypercube is connected, so Δ is irreducible. Adding the diagonal $\hat{V}_{H,n}$ does not change the off-diagonal pattern, so $H_{H,n}$ is irreducible. Consider the shifted matrix

$$M_{\text{shift}} = \alpha I - H_{H,n}, \quad \alpha = M^2 + 2M + 1.$$

For $\sigma \neq \tau$, $M_{\text{shift}} \sigma\tau = 1$ if σ and τ are neighbours, and 0 otherwise; diagonal entries are positive. Hence M_{shift} is non-negative and irreducible. By the Perron-Frobenius theorem, it has a simple largest eigenvalue $\rho(M_{\text{shift}})$ with a strictly positive eigenvector v . Then

$$H_{H,n}v = (\alpha - \rho(M_{\text{shift}}))v,$$

so v is an eigenvector of $H_{H,n}$ for the smallest eigenvalue λ_0 . Normalising gives the ground state ψ_0 with $\psi_0(\sigma) > 0$ for all σ .

Theorem 4.1 (P1 for Erdős–Hajnal Hamiltonian). *$H_{H,n}$ has a unique strictly positive ground state ψ_0 .*

4.2 Pillar P2 – Constant spectral gap

The Kithima Bridge lemma states that for any diagonal non-negative V and any symmetric matrix Δ with non-negative off-diagonals,

$$\gamma(V + \Delta) \geq \gamma(\Delta).$$

Here $\gamma(\Delta) = 2$. Since $\hat{V}_{H,n}$ is diagonal and non-negative, we obtain

$$\gamma(H_{H,n}) \geq 2.$$

Theorem 4.2 (P2 for Erdős–Hajnal Hamiltonian). *The spectral gap satisfies $\gamma(H_{H,n}) \geq 2$.*

4.3 Pillar P3 – Logarithmic area law and MPS representation

The constant spectral gap implies exponential decay of correlations. By the Brandão–Horodecki theorem and a Hilbert curve ordering, the entanglement entropy satisfies $S(\rho_B) = O(\log M)$. Hence the maximum Schmidt rank is at most $e^{O(\log M)} = M^{O(1)}$. Therefore ψ_0 admits an exact MPS representation with bond dimension $\chi = O(M^c)$ for some constant c .

Theorem 4.3 (P3 for Erdős–Hajnal Hamiltonian). *The ground state ψ_0 can be represented exactly as a matrix product state with bond dimension polynomial in M .*

4.4 Pillar P4 – Deterministic extraction

The power iteration algorithm on MPS (Article I.5) converges to ψ_0 in $O(\log M)$ steps because the spectral gap is constant. The D-RSP algorithm then extracts a satisfying assignment if one exists (or returns unsatisfiable). The algorithm runs in time polynomial in M .

Theorem 4.4 (P4 for Erdős–Hajnal Hamiltonian). *There exists a deterministic polynomial-time algorithm that, given the Hamiltonian $H_{H,n}$, decides the satisfiability of $\Phi_{H,n}$ and, if satisfiable, returns a satisfying assignment (i.e., a counterexample graph G).*

5 Application to the Erdős–Hajnal conjecture

For each fixed H and each $n < n_0(H)$, we construct $H_{H,n}$ as above. The D-RSP algorithm decides whether $\Phi_{H,n}$ is satisfiable. In Article XXVI.4 we will run the solver for all such n and prove that $\Phi_{H,n}$ is unsatisfiable for every $n < n_0(H)$. Since the asymptotic bound (XXVI.1) guarantees that the property holds for all $n \geq n_0(H)$, the unsatisfiability of these finitely many SAT instances proves the Erdős–Hajnal conjecture for the fixed H . Since H was arbitrary, the conjecture holds for all finite graphs.

6 Formalisation in Lean4

The Lean formalisation imports the generic theorems from the kernel and instantiates them with the SAT instance $\Phi_{H,n}$.

Listing 1: SeriesXXVI/ErdosHajnalHamiltonian.lean (final)

```

1 import XXVI.ErdosHajnalCircuit
2 import Tseitin
3 import SpectralLibrary
4 import KithimaBridge
5 import MPSAreaLaw
6 import PowerIteration
7
8 open SpectralLibrary
9
10 variable (H : SimpleGraph) [Fintype V(H)] (n : ℕ) (hn : n < n₀ H)
11   threshold H)
12
13 def C : Circuit (Fin (n*(n-1)/2)) Bool := erdos_hajnal_circuit H n
14 def SATInstance := tseitin C
15 def M : Matrix (Fin M) (Fin M) := adjacencyMatrixOf (hypercube_graph M)
16 def Config := Fin M Bool
17
18 def V (C : Config) : Bool := if satisfies C then 0 else M^2
19 def H_Hn : Matrix Config Config := diagonal V +
20
21 lemma H_irreducible : Irreducible H_Hn :=
22   matrix_is_irreducible_of_adjacency_graph (hypercube_connected M)
23
24 theorem ground_state_unique_pos :
25   ! (C : Config) (M : ℕ) (M > 0) (M < 2)
26   (H_Hn * C) = ( _min H_Hn ) :=
27   perron_frobenius H_Hn

```

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28
29 theorem spectral_gap_constant : spectral_gap H_Hn      2 :=
30   kithima_bridge (diagonal V) (by simp [V, M])      (by simp) 1 (by
31     norm_num)
32
33 theorem area_law :      C,      B, entanglement_entropy (ground_state H_Hn
34   ) B      C * Real.log M :=
35   renormalisation_area_law (by sorry) (hilbert_curve_bound (clause_graph
36     ) (by exact bounded_degree))
37
38 theorem drsp_algorithm :      alg, polynomial_time alg
39   (      , satisfies      alg returns      ) :=
40   drsp_correct H_Hn

```

All theorems are proved in the library; the file only instantiates them.

7 Conclusion

We have constructed the spectral Hamiltonian for the SAT instance encoding the existence of a counterexample to the Erdős–Hajnal conjecture for fixed H and fixed $n < n_0(H)$. The four pillars are verified, so the spectral SAT solver applies. The solver will decide satisfiability in polynomial time. The next article (XXVI.4) will run this decision for all such n and prove that no counterexample exists, thereby confirming the conjecture.

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